

INDIAN STATISTICAL INSTITUTE

Parametric Inference

(B.Stat. 3rd year)

Semester 1, 2026–27

Homework 3

1. Let

$$X_1, X_2, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \frac{1}{\theta} e^{-x/\theta}, \quad 0 < x, \theta < \infty.$$

Derive the form of the UMP test for testing $H_0 : \theta \leq 1$ vs. $H_A : \theta > 1$ with a given size $0 < \alpha < 1$.

2. Let

$$X_1, X_2, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \frac{\theta^x}{x!} e^{-\theta}, \quad 0 < \theta < \infty, \quad x = 0, 1, \dots$$

Derive the form of the UMP test for testing $H_0 : \theta \leq 2$ vs. $H_A : \theta > 2$ with a given size $0 < \alpha < 1$.

3.

$$\begin{aligned} H_0 : X &\sim p_0(x) = \left(\frac{1}{2}\right)^{x+1}, & x = 0, 1, 2, \dots \\ H_A : X &\sim p_1(x) = \frac{1}{x!} e^{-1}, & x = 0, 1, 2, \dots \end{aligned}$$

Derive the form of the MP test for testing H_0 vs. H_A with a given size $0 < \alpha < 1$.

4. $X \sim \text{Uniform}[0, \theta]$. Derive the form of the UMP test for $H_0 : \theta = 1$ vs. $H_A : \theta \neq 1$ with a given size $0 < \alpha < 1$.